

Notes: Stavrakeva & Tang (JF 2026)

The Dollar during the Great Recession: The Information Channel of U.S. Monetary Policy and the “Flight to Safety”

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1 Executive summary

1.1 What is the puzzle?

Standard open-economy logic suggests that *expansionary* U.S. monetary policy (rate cuts / easing surprises) should lower U.S. yields and (other things equal) **depreciate** the dollar. During the Great Recession (December 16, 2008–September 13, 2012), the paper documents the opposite at medium horizons:

*U.S. monetary policy easings were followed by a **dollar appreciation** three to four weeks later.*

1.2 Core mechanism

The paper argues that during the Great Recession the **information channel** dominated the direct interest-rate channel: an easing surprise was interpreted as the Fed revealing *worse future fundamentals*, which raised effective risk aversion, triggered a **flight to safety into dollars**, and thus strengthened the dollar.

1.3 Key empirical signature

The central signature is a **horizon-dependent sign reversal**:

- **Short horizon (same day / first weeks)**: conventional sign (easing $\Rightarrow$ weaker dollar, stronger equities).
- **Medium horizon (3–8 weeks)**: reversal in the Great Recession (easing $\Rightarrow$ stronger dollar, weaker equities).

1.4 What matters most for the reversal

Using two ways to separate forward guidance (FG) from quantitative easing (QE/LSAP), the paper concludes that:

- the reversal is **driven mainly by forward guidance** (non-QE days / FG factor),
- and it is **stronger against “bad hedge” currencies** (those that tend to depreciate in global bad times).

2 Empirical design and identification

2.1 High-frequency monetary policy surprises

The paper uses a standard identification strategy in monetary event studies:

- Focus on FOMC announcement dates (plus QE announcements outside regular statements),

- Construct the monetary surprise as the **change in the implied interest rate** from Eurodollar futures expiring three quarters ahead (ED4) measured in a tight one-hour window around the announcement.

This surprise isolates *unexpected* policy news at high frequency.

2.2 Sample splits

The analysis emphasizes three subsamples:

- Pre–Great Recession: Jan 1990–Nov 2008.
- Great Recession: Dec 16, 2008–Sep 13, 2012.
- Post–Great Recession: Sep 2012–Dec 2019 (ending before COVID to avoid a new regime).

2.3 Currency sets

The exchange-rate outcomes are averages across:

- 9 advanced-economy currencies (AUD, CAD, CHF, EUR/DEM, GBP, JPY, NOK, NZD, SEK),
- 15 emerging-market currencies (BRL, CLP, COP, CZK, ILS, INR, ISK, KRW, MXN, PHP, RUB, SGD, THB, TRY, ZAR),

excluding pegged periods.

3 Model block I: local projections for impulse responses

3.1 Exchange-rate impulse responses (Equation (1))

For each horizon $n = 0, \dots, 90$ days, the paper estimates:

$$\bar{s}_{\tau+n} - \bar{s}_{\tau-1} = \alpha_n + \beta_n^{\Delta s} mp_\tau + \gamma_n (\bar{s}_{\tau-1} - \bar{s}_{\tau-2}) + \varepsilon_{\tau,n}. \quad (1)$$

Definitions.

- τ indexes monetary policy announcement dates.
- mp_τ is the ED4 implied-rate surprise over the one-hour announcement window.
- $s_{k,t}$ is the bilateral nominal exchange rate (units of currency k per USD); $\bar{s}_t \equiv \frac{1}{N} \sum_{k=1}^N s_{k,t}$ is the cross-currency average.

Sign interpretation (important). Because mp_τ is an *interest-rate* surprise, an **easing** corresponds to $mp_\tau < 0$. Hence:

- $\beta_n^{\Delta s} > 0$: an easing ($mp_\tau < 0$) tends to *lower* $\bar{s} \Rightarrow$ **dollar depreciation**.
- $\beta_n^{\Delta s} < 0$: an easing tends to *raise* $\bar{s} \Rightarrow$ **dollar appreciation**.

Standard errors are Newey–West to handle overlapping horizons.

3.2 Cross-sectional heterogeneity: hedge-quality interactions (Equation (2))

To quantify how the response varies across currencies, the paper estimates:

$$s_{k,\tau+n} - s_{k,\tau-1} = \alpha_{k,n} + \beta_n^{\Delta s} mp_\tau + \beta_n^{hedge} (hedge_k \times mp_\tau) + \gamma_{n,k} (s_{k,\tau-1} - s_{k,\tau-2}) + \varepsilon_{k,\tau,n}. \quad (2)$$

Interpretation. β_n^{hedge} measures how the impulse response differs with a currency’s *hedging quality* $hedge_k$. A higher $hedge_k$ means the currency tends to *appreciate* in bad times (better hedge). The paper uses three proxies for $hedge_k$:

- covariance with the S&P 500 (“bad times” $\approx$ equity drawdowns),
- covariance with intermediary leverage/capital (He–Kelly–Manela proxy),
- average interest-rate differential (carry/funding-currency proxy).

4 Empirical results: what Figures 1–6 and Tables I–III establish

4.1 The baseline impulse responses (Figure 1; Tables I–II)

Main fact. In pre- and post-Great-Recession samples, the impulse responses match conventional wisdom: tightening surprises ($mp_\tau > 0$) are followed by a **dollar appreciation**. In the Great Recession, this relationship **reverses at medium horizons**: after about 3–4 weeks, easings ($mp_\tau < 0$) are followed by a **dollar appreciation**.

Why the dashed lines matter. Figure 1 and Tables I–II also report estimates excluding major QE announcement surprises during the Great Recession. The sign reversal becomes **stronger** when focusing on the “non-QE” subsample, suggesting the reversal is tied to FG-type news rather than asset purchases.

4.2 Separating FG vs LSAP using Swanson (2021) factors (Figure 2; Table III)

A second identification strategy uses Swanson’s (2021) high-frequency factors that separately capture:

- forward guidance (FG) surprises,
- LSAP (QE) surprises.

During the Great Recession, the FG factor reproduces the **sign reversal** at medium horizons, while LSAP shocks generate more conventional patterns (no strong reversal), reinforcing the conclusion that the Great-Recession anomaly is mainly FG-driven.

4.3 Cross-currency heterogeneity (Figures 3–4)

Key cross-sectional pattern. Currencies that are **worse hedges** (tend to depreciate in global bad times) experience larger depreciations against the USD following U.S. easing news during the Great Recession, consistent with a **flight-to-safety into dollars**.

5 Model block II: a parsimonious information-channel theory

5.1 U.S. macro block (Equations (3)–(5))

The paper posits a simple partial-equilibrium daily U.S. block:

$$\pi_t^{us} = \alpha y_t^{us}, \quad (3)$$

$$y_t^{us} = -\nu(i_t^{us} - \pi_t^{us}) + \varepsilon_t^{y,us}, \quad (4)$$

with $\alpha, \nu > 0$ and $\varepsilon_t^{y,us} \sim \mathcal{N}(0, \sigma_y^2)$.

Monetary policy follows a Taylor rule with a monetary shock:

$$i_t^{us} = \phi_y y_t^{us} + \phi_\pi \pi_t^{us} + \varepsilon_t^{mp,us}, \quad \varepsilon_t^{mp,us} \sim \mathcal{N}(0, \sigma_{mp}^2),$$

uncorrelated with $\varepsilon_t^{y,us}$.

Solving yields (write $\kappa \equiv \phi_y + \alpha(\phi_\pi - 1)$ and $\eta \equiv 1 - \nu\alpha$):

$$\begin{aligned} y_t^{us} &= \frac{\varepsilon_t^{y,us} - \nu \varepsilon_t^{mp,us}}{1 + \nu\kappa}, & \pi_t^{us} &= \frac{\alpha \varepsilon_t^{y,us} - \nu \varepsilon_t^{mp,us}}{1 + \nu\kappa}, \\ i_t^{us} &= \frac{(\kappa + \alpha) \varepsilon_t^{y,us} + \eta \varepsilon_t^{mp,us}}{1 + \nu\kappa}, & r_t^{us} &= \frac{\kappa \varepsilon_t^{y,us} + \varepsilon_t^{mp,us}}{1 + \nu\kappa}. \end{aligned} \quad (5)$$

5.2 Asset-pricing block for exchange rates and equities (Equations (6)–(7))

Let $SDF_{t,t+1}$ be the real stochastic discount factor of the marginal trader and π_{t+1} inflation in her consumption basket. Currency- i and USD bonds imply:

$$\mathbb{E}_t \left[SDF_{t,t+1} e^{-\pi_{t+1}} \left((1 + i_t^{us}) - \frac{S_t}{S_{t+1}} (1 + i_t^i) \right) \right] = 0. \quad (6)$$

Equities satisfy:

$$\mathbb{E}_t \left[SDF_{t,t+1} e^{-\pi_{t+1}} \left(e^{r_{t+1}^{eq}} - (1 + i_t^{us}) \right) \right] = 0. \quad (7)$$

5.3 Risk premia and “hedge” currencies (Equations (8)–(13))

Assuming joint log-normality, the expected excess currency return (long USD bond, short foreign bond) can be written as:

$$\lambda_t \equiv \mathbb{E}_t[\Delta s_{t+1}] - \tilde{i}_t = \frac{\text{Var}_t(\Delta s_{t+1})}{2} - \text{Cov}_t(sdf_{t,t+1} - \pi_{t+1}, \Delta s_{t+1}), \quad (8)$$

where $\tilde{i}_t \equiv i_t^i - i_t^{us}$ and $sdf_{t,t+1} \equiv \log SDF_{t,t+1}$. Similarly, the equity premium:

$$\lambda_t^{eq} \equiv \mathbb{E}_t[r_{t+1}^{eq}] - i_t^{us} = -\frac{\text{Var}_t(r_{t+1}^{eq})}{2} - \text{Cov}_t(sdf_{t,t+1} - \pi_{t+1}, r_{t+1}^{eq}). \quad (9)$$

With CRRA and constant second moments, the premia are linear in time-varying risk aversion

ρ_t :

$$\lambda_t = \frac{\sigma_s^2}{2} + \sigma_{\pi,s} + \rho_t \sigma_{c,s}, \quad (10)$$

$$\lambda_t^{eq} = -\frac{\sigma_{eq}^2}{2} + \sigma_{\pi,eq} + \rho_t \sigma_{c,eq}, \quad (11)$$

with ρ_t countercyclical:

$$\rho_t = \rho_y y_t^{us}, \quad \rho_y < 0. \quad (12)$$

A useful long-run cross-sectional implication is the time-average relation:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}_t[\Delta s_{t+1}] - \frac{1}{T} \sum_{t=1}^T \tilde{i}_t = \frac{\sigma_s^2}{2} + \frac{1}{T} \sum_{t=1}^T \rho_t \sigma_{c,s} + \sigma_{\pi,s}. \quad (13)$$

This provides theoretical support for using interest-rate differentials as a proxy for hedging properties.

6 Model block III: forward guidance as information + misperceived signal precision

6.1 Forward guidance announcement perceived with bias (Equation (14))

The paper models an FG announcement at $t+1$ as the Fed truthfully announcing its expected policy rate in a future period $t+h$. Investors face a signal-extraction problem and misperceive the precision of the Fed's private signal. Their perceived data-generating process for the announcement is:

$$\tilde{a}_{t+l} = \frac{\kappa + \alpha}{1 + \nu\kappa} \frac{1}{\sigma_{\tilde{\varepsilon},CB}^2 + q_{t+l}^{CB}} \left(\frac{1}{\sigma_y^2} + \frac{1}{\sigma_{\tilde{\varepsilon},CB}^2 + q_{t+l}^{CB}} \right)^{-1} \tilde{\varepsilon}_{t+1}^{y,us,CB} + \frac{\eta}{1 + \nu\kappa} \varepsilon_{t+h}^{mp,us}, \quad (14)$$

where $q_{t+l}^{CB} \geq 0$ captures investors thinking the central bank's signal is noisier than it truly is (before minutes are released).

6.2 Revisions in output expectations (Equations (15)–(16))

Investor revisions of expected future output satisfy:

$$\mathbb{E}_{t+l}[y_{t+h}^{us}] - \mathbb{E}_t[y_{t+h}^{us}] = \mathbb{E}_{t+l}[y_{t+h}^{us}] = K_{t+l} a_{t+1}, \quad 1 \leq l < h, \quad (15)$$

where

$$K_{t+l} \equiv \frac{\frac{(\kappa + \alpha)\sigma_y^2}{\sigma_{mp}^2} - \left(\frac{\sigma_{\tilde{\varepsilon},CB}^2 + q_{t+l}^{CB}}{\sigma_y^2} + 1 \right) \nu\eta}{\frac{(\kappa + \alpha)^2\sigma_y^2}{\sigma_{mp}^2} + \left(\frac{\sigma_{\tilde{\varepsilon},CB}^2 + q_{t+l}^{CB}}{\sigma_y^2} + 1 \right) \eta^2}. \quad (16)$$

Economic meaning of K_{t+l} .

- If $K_{t+l} < 0$: the **direct** effect dominates (lower expected rates $\Rightarrow$ higher expected output).

- If $K_{t+l} > 0$: the **information** effect dominates (lower expected rates $\Rightarrow$ worse expected fundamentals).

The paper argues K_{t+l} was more likely positive during the Great Recession, especially once investors learn the Fed's signal is more precise.

7 Model block IV: exchange-rate and equity decompositions

7.1 Exchange-rate change decomposition (Equations (17)–(21))

Iterating forward gives an exchange-rate level expression and a change decomposition:

$$\Delta s_{t+l,t} = \underbrace{\sum_{k=0}^{l-1} \mathbb{E}_t \tilde{i}_{t+k} - \sum_{k=l}^{\infty} (\mathbb{E}_{t+l} - \mathbb{E}_t) \tilde{i}_{t+k}}_{\phi_{t+l,t}^{EH}} + \underbrace{\sum_{k=0}^{l-1} \mathbb{E}_t \lambda_{t+k} - \sum_{k=l}^{\infty} (\mathbb{E}_{t+l} - \mathbb{E}_t) \lambda_{t+k}}_{\phi_{t+l,t}^{\lambda}} + \underbrace{\lim_{K \rightarrow \infty} (\mathbb{E}_{t+l} s_{t+K} - \mathbb{E}_t s_{t+K})}_{\phi_{t+l,t}^{LR}}. \quad (17)$$

The long-run component can be expressed in terms of relative inflation (trend-stationary real exchange rate):

$$\phi_{t+l,t}^{LR} = \lim_{K \rightarrow \infty} \left(\mathbb{E}_{t+l} [q_{t+l+K} + \tilde{p}_{t+l+K}] - \mathbb{E}_t [q_{t+l+K} + \tilde{p}_{t+l+K}] \right), \quad (18)$$

$$= \sum_{k=1}^{\infty} (\mathbb{E}_{t+l} \tilde{\pi}_{t+l+k} - \mathbb{E}_t \tilde{\pi}_{t+k}) + (\tilde{p}_{t+l} - \tilde{p}_t). \quad (19)$$

A key derivative for the risk-premium component is:

$$\frac{\partial \phi_{t+l,t}^{\lambda}}{\partial a_{t+1}} = -\frac{\partial \mathbb{E}_{t+l} \lambda_{t+h}}{\partial a_{t+1}} = -\rho_y \sigma_{c,s} K_{t+l}. \quad (20)$$

Putting the pieces together yields the overall derivative:

$$\frac{\partial \Delta s_{t+l,t}}{\partial a_{t+1}} = -\frac{\partial}{\partial a_{t+1}} \sum_{k=l}^{\infty} \mathbb{E}_{t+l} \tilde{i}_{t+k}^i + \frac{\partial}{\partial a_{t+1}} \sum_{k=1}^{\infty} \mathbb{E}_{t+l} \pi_{t+l+k}^i + 1 - (\sigma_{c,s} \rho_y + \alpha) K_{t+l}. \quad (21)$$

Intuition. The nominal-rate-expectations channel is relatively stable across regimes. The sign reversal during the Great Recession arises because $K_{t+l} > 0$ makes risk-premium and inflation-expectation revisions dominate, generating flight-to-safety dollar strength after an easing announcement.

7.2 Equity decomposition via Campbell–Shiller (Equation (22))

Using Campbell–Shiller, the equity premium can be written as:

$$\lambda_t^{eq} = \mathbb{E}_t[r_{t+1}^{eq}] - i_t^{us} = \tilde{k} + \tilde{\rho} \mathbb{E}_t[pd_{t+1} - pd_t] + \tilde{\rho} \mathbb{E}_t[\Delta d_{t+1}] - i_t^{us}, \quad (22)$$

where $pd_t \equiv p_t^{eq} - d_t$ is the log price-dividend ratio.

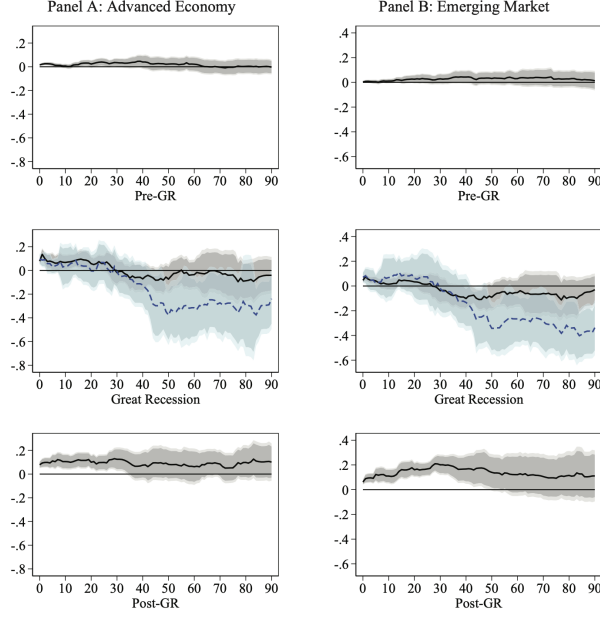


Figure 1. Impulse responses of exchange rate to U.S. monetary policy surprises.

7.3 Empirical decompositions of impulse responses (Equations (23)–(24))

Empirically, the paper constructs a daily measure of the *nominal-rate-expectations component* using a VAR disciplined by survey forecasts. The remainder is treated as a residual (risk premia + inflation for FX; risk premia + dividend growth for equities). This yields:

$$\beta_n^{\Delta s} = \beta_n^{\phi^{EH}} + \beta_n^{\phi^{\lambda} + \phi^{LR}}, \quad (23)$$

$$\beta_n^{\Delta p^{eq}} = \beta_n^{\phi^{EH,eq}} + \beta_n^{\phi^{\lambda,eq} + \phi^{D,eq}}. \quad (24)$$

Main empirical conclusion from the decompositions. The Great-Recession sign reversal is **not** driven by the nominal-rate-expectations component. It is driven by the residual component:

- FX: expected excess currency returns and/or relative inflation expectations,
- equities: expected excess equity returns and/or dividend-growth expectations.

8 Interpreting each figure (1–9) and each table (I–IV)

8.1 Figure 1 (baseline FX impulse responses)

What it is. Impulse responses of average FX changes to ED4 surprises across horizons $n = 0, \dots, 90$. Panels: advanced economies vs emerging markets. Subpanels: pre-GR, GR, post-GR. Solid lines include all policy dates; dashed lines exclude QE announcements in the GR subsample.

What to learn.

- Pre- and post-GR: conventional sign (tightening $\Rightarrow$ USD up).
- GR: after ≈ 20 –30 days, sign flips (easing $\Rightarrow$ USD up), especially excluding QE dates.

8.2 Tables I–II (FX impulse responses at selected horizons)

What they do. Tabulate the same impulse responses as Figure 1 at horizons 0, 10, . . . , 90 days.

How to read the sign. A negative coefficient at $n \geq 30$ during GR means:

$$mp_{\tau} < 0 \text{ (easing)} \quad \Rightarrow \quad \Delta s > 0 \text{ (USD appreciates).}$$

Table I
Impulse Responses of Advanced Economy Exchange Rate Index to U.S. Monetary Policy Surprises

This table presents responses of the log exchange rate to the change in Eurodollar futures expiring three quarters hence (ED4) over a one-hour window around FOMC announcements. Estimates are multiplied by 100 to facilitate comparison. Newey-West standard errors are in parentheses. Estimates are based on average log changes of the AUD, CAD, CHF, EUR (DEM prior to 1999), GBP, JPY, NOK, NZD, and SEK. The full sample is from 1990 through 2019, excluding periods of pegging for each currency. The Great Recession subsample comprises the policy announcements from December 16, 2008 through September 13, 2012. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Horizon (Days)	Pre-GR		Great Recession				Post-GR	
			All Policy Shocks		Excluding QE Shocks			
	173 Obs		33 Obs		19 Obs		60 Obs	
	Coefficient	R ²	Coefficient	R ²	Coefficient	R ²	Coefficient	R ²
0	1.56*** (0.37)	0.13	7.85*** (1.56)	0.56	9.61*** (2.20)	0.48	7.82*** (1.39)	0.38
10	0.67 (0.97)	0.01	6.71* (3.40)	0.10	3.93 (8.62)	0.05	10.54*** (3.29)	0.20
20	2.30 (1.67)	0.01	5.65* (3.08)	0.09	−0.43 (7.54)	0.09	9.99*** (3.10)	0.11
30	3.46* (1.93)	0.02	−1.87 (5.62)	0.02	−1.04 (7.58)	0.14	12.48** (5.00)	0.12
40	4.02 (2.97)	0.02	−7.13 (5.80)	0.06	−12.26 (8.97)	0.29	7.53 (6.24)	0.05
50	2.26 (2.89)	0.00	−7.36 (6.07)	0.07	−37.68*** (11.89)	0.42	8.11 (7.13)	0.05
60	2.13 (3.28)	0.00	−3.56 (6.19)	0.04	−31.48** (13.97)	0.32	6.22 (7.65)	0.04
70	−0.38 (3.04)	0.00	−1.57 (9.05)	0.01	−27.32* (14.32)	0.27	8.48 (6.09)	0.04
80	0.34 (3.29)	0.00	−8.74 (8.61)	0.03	−29.57 (20.17)	0.20	9.83 (7.33)	0.05
90	−0.31 (3.09)	0.00	−4.10 (7.57)	0.01	−23.95* (12.00)	0.16	10.12 (7.84)	0.03

Table II
Impulse Responses of Emerging Market Exchange Rate Index to U.S. Monetary Policy Surprises

This table presents responses of the log exchange rate to the change in Eurodollar futures expiring three quarters hence (ED4) over a one-hour window around FOMC announcements. Estimates are multiplied by 100 to facilitate comparison. Newey-West standard errors are in parentheses. Estimates are based on average log changes of the BRL, CLP, COP, CZK, ILS, INR, ISK, KRW, MXN, PHP, RUB, SGD, THB, TRY, and ZAR. Averages are calculated excluding periods of pegging for each currency. The full sample is from 1990 through 2019. The Great Recession subsample comprises the policy announcements from December 16, 2008 through September 13, 2012. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Horizon (Days)	Pre-GR		Great Recession				Post-GR	
			All Policy Shocks		Excluding QE Shocks			
	173 Obs		33 Obs		19 Obs		60 Obs	
	Coefficient	R ²	Coefficient	R ²	Coefficient	R ²	Coefficient	R ²
0	0.35 (0.37)	0.04	4.63*** (0.92)	0.41	7.00*** (2.06)	0.37	6.12*** (1.01)	0.37
10	1.06 (0.94)	0.02	1.43 (2.41)	0.07	6.90 (8.79)	0.22	11.03*** (2.97)	0.16
20	2.54 (1.79)	0.06	3.33 (2.62)	0.09	6.21 (8.76)	0.12	16.12*** (2.94)	0.22
30	2.84 (2.23)	0.05	−3.80 (5.56)	0.04	−1.98 (10.34)	0.15	20.09*** (4.09)	0.22
40	4.29 (2.76)	0.06	−10.12** (4.49)	0.10	−13.29 (10.88)	0.24	16.69*** (5.98)	0.12
50	3.12 (2.71)	0.04	−9.12 (5.42)	0.11	−34.27*** (10.18)	0.34	13.99* (7.38)	0.06
60	3.03 (3.00)	0.04	−5.25 (5.74)	0.08	−26.58** (10.54)	0.25	12.16 (8.65)	0.04
70	3.71 (3.80)	0.04	−6.01 (8.74)	0.04	−28.67** (12.89)	0.21	10.58 (9.19)	0.03
80	2.74 (3.52)	0.03	−9.07 (8.22)	0.04	−34.02** (12.20)	0.14	11.52 (9.86)	0.03
90	1.32 (3.99)	0.03	−2.95 (6.38)	0.01	−33.72*** (10.53)	0.11	11.13 (10.29)	0.02

Table I (advanced economies).

- Same-day $n = 0$: positive and significant across subsamples (conventional immediate reaction).
- GR: coefficients turn negative from about $n = 30$ onward, and become very negative (and significant) in the non-QE subsample at $n \in [50, 90]$.

Table II (emerging markets). Qualitatively similar to Table I, with an even stronger medium-horizon reversal in the non-QE subsample.

8.3 Figure 2 and Table III (Swanson FG vs LSAP factors in GR)

What they show. Impulse responses in the Great Recession using separate FG and LSAP surprise measures.

Main message.

- FG shocks reproduce the GR sign reversal (short-horizon conventional, medium-horizon flipped).
- LSAP shocks do not generate the same reversal pattern, implying the GR anomaly is largely FG/information-driven.

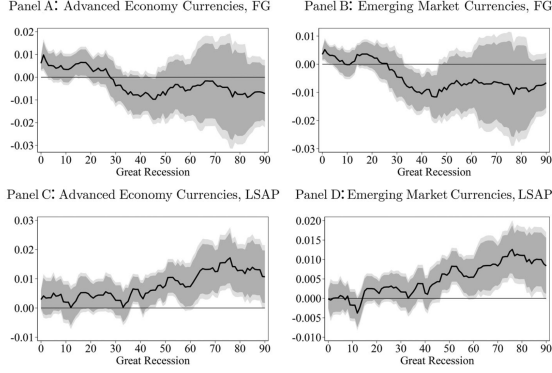


Figure 2. Impulse responses of exchange rates during the Great Recession to FG and LSAP factors from Swanson (2021).

Table III
Impulse Responses of Exchange Rates during the Great Recession to FG and LSAP Factors from Swanson (2021)

This table presents responses of the log exchange rate to the LSAP and FG factors from Swanson (2021). Estimates are multiplied by 100 to facilitate comparison. Newey-West standard errors are in parentheses. The LSAP factor is negated so that a surprise increase in asset purchases corresponds to a positive surprise in interest rate futures. The advanced economy estimates are based on average log changes of the AUD, CAD, CHF, EUR (DEM prior to 1999), GBP, JPY, NOK, NZD, and SEK. The emerging market estimates are based on average log changes of the BRL, CLP, COP, CZK, ILS, INR, ISK, KRW, MXN, PHP, RUB, SGD, THB, TRY, and ZAR. Averages are calculated excluding periods of pegging for each currency. The full sample is July 5, 1991 through June 19, 2019. The Great Recession subsample comprises the policy announcements from December 16, 2008 through September 13, 2012. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Horizon (Days)	Advanced Economies				Emerging Markets			
	Fwd Guidance		LSAP		Fwd Guidance		LSAP	
	30 Obs		30 Obs		30 Obs		30 Obs	
	Coefficient	R ²	Coefficient	R ²	Coefficient	R ²	Coefficient	R ²
0	0.61*** (0.17)	0.47	0.29 (0.18)	0.16	0.35*** (0.11)	0.35	-0.01 (0.14)	0.05
10	0.34 (0.34)	0.04	0.20 (0.35)	0.02	-0.01 (0.25)	0.07	-0.18 (0.20)	0.08
20	0.39 (0.31)	0.06	0.40 (0.28)	0.08	0.28 (0.25)	0.09	0.19 (0.18)	0.08
30	-0.38 (0.45)	0.03	0.26 (0.30)	0.03	-0.45 (0.44)	0.07	0.17 (0.24)	0.04
40	-0.85* (0.47)	0.11	0.43 (0.25)	0.06	-1.05*** (0.34)	0.18	0.14 (0.22)	0.04
50	-0.79 (0.62)	0.10	0.77** (0.31)	0.12	-0.91 (0.55)	0.15	0.66*** (0.24)	0.13
60	-0.56 (0.67)	0.06	0.73* (0.40)	0.10	-0.66 (0.53)	0.12	0.55* (0.28)	0.11
70	-0.30 (0.90)	0.02	1.24*** (0.43)	0.17	-0.64 (0.77)	0.06	0.84*** (0.30)	0.11
80	-0.76 (1.07)	0.03	1.29*** (0.44)	0.14	-0.84 (0.89)	0.05	1.11*** (0.36)	0.12
90	-0.72 (0.72)	0.03	1.07* (0.53)	0.09	-0.66 (0.52)	0.03	0.84* (0.42)	0.07

8.4 Figure 3 (cross-sectional FX response vs hedge quality)

What it is. A scatter relating currency-specific 90-day responses (non-QE GR shocks) to hedge-quality measures.

Main message. “Bad hedge” currencies move more against the USD after easing news, consistent with a flight to safety into USD.

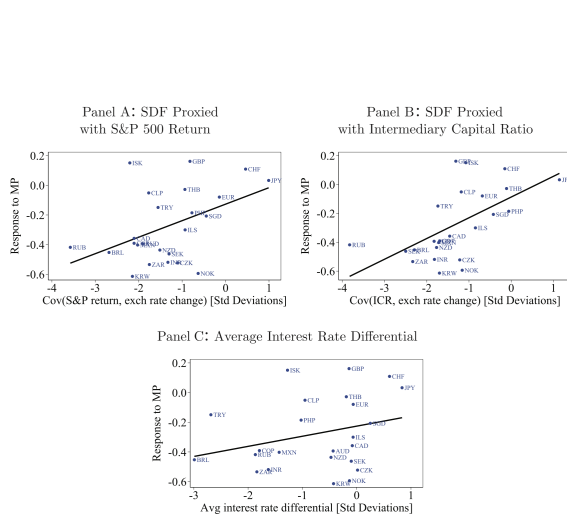


Figure 3. Currency-specific response of exchange rate changes to non-QE U.S. monetary policy surprises during the Great Recession versus hedging quality.

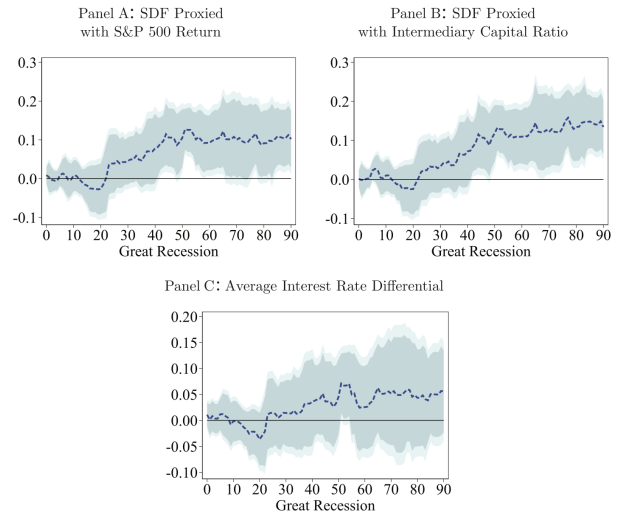


Figure 4. Impulse responses of exchange rates to U.S. monetary policy surprises interacted with hedging quality (Great Recession subsample).

8.5 Figure 4 (panel interaction: response varies with hedge quality)

What it is. Panel impulse responses allowing the effect to vary with $hedge_k$ (Equation (2)).

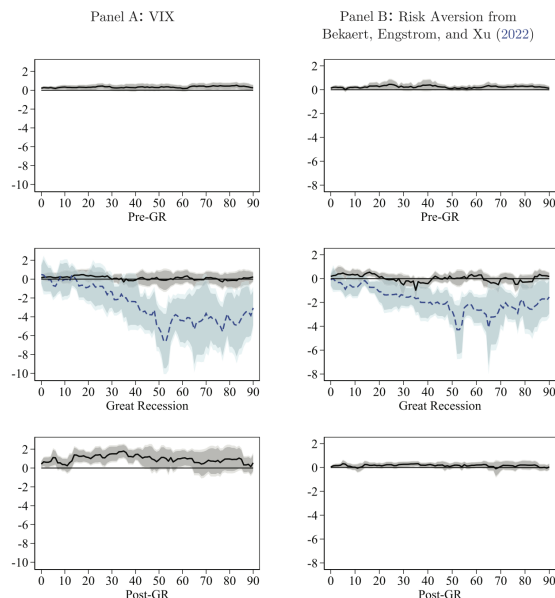


Figure 7. Impulse responses of risk aversion proxies to U.S. monetary policy surprises.

Main message. The GR reversal is strongest for currencies with worse hedge properties.

8.6 Figures 5–6 (equity price impulse responses)

What they show. Impulse responses of the S&P 500 and MSCI World indices to U.S. monetary surprises. Outside the Great Recession, tightenings are followed by equity declines (conventional). During the Great Recession, the medium-horizon responses reflect the information channel: easing news is followed by equity *declines* at medium horizons (worse fundamentals / higher premia).

8.7 Figure 7 (risk aversion responses)

What it is. Impulse responses of VIX and a risk-aversion index to monetary surprises.

Main message. During the Great Recession, forward-guidance easing news increases risk aversion at horizons of roughly 3–8 weeks, matching the “information $\Rightarrow$ flight to safety” mechanism.

8.8 Figures 8–9 (component decompositions)

What they show. Decompositions of FX and equity responses into:

- nominal-rate-expectations component,
- residual component (risk premia + inflation for FX; risk premia + dividend growth for equities).

Main message. The Great-Recession structural break is concentrated in the residual component, not in the nominal-rate-expectations component.

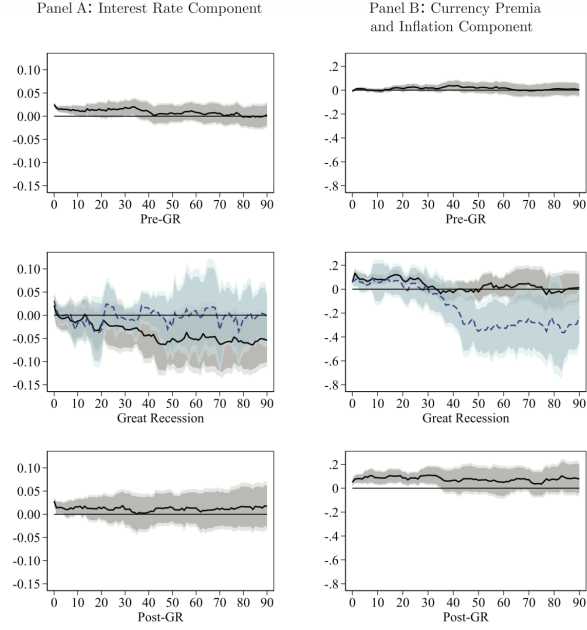


Figure 8. Impulse responses of **exchange rate components** to U.S. monetary policy surprises.

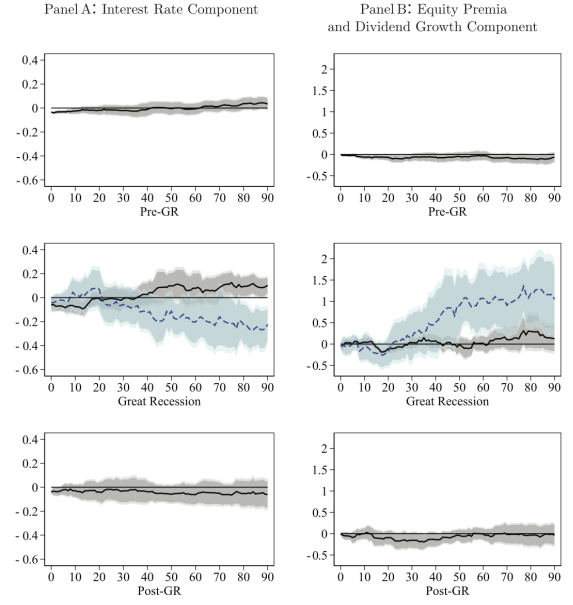


Figure 9. Impulse responses of **equity price components** to U.S. monetary policy surprises.

Table IV
Subsample Means of Uncertainty Measures

The JLN macro uncertainty measure is the 12-month-ahead macroeconomic uncertainty estimated of Jurado, Ludvigson, and Ng (2015). GDP forecast dispersion is the 25th to 75th percentile range of four-quarter-ahead U.S. real GDP forecasts from *Blue Chip Financial Forecasts*. BBD monetary policy uncertainty is the monetary policy component of the Baker, Bloom, and Davis (2016) policy uncertainty index. All three measures are standardized over the full 1990 through 2019 sample to facilitate interpretation.

	JLN Macro Uncertainty	GDP Forecast Dispersion	BBD Monetary Policy Uncertainty
Jan 1990–Nov 2008	0.07	0.13	0.16
Dec 2008–Aug 2012	0.68	0.86	−0.02
Sep 2012–Dec 2019	−0.54	−0.80	−0.43

8.9 Table IV (why the Great Recession was special)

What it shows. Subsample means of macro uncertainty measures vs monetary policy uncertainty. Macro uncertainty is **much higher** during the Great Recession, while monetary policy uncertainty is not. This supports the model condition that the information channel dominates when uncertainty about fundamentals is high relative to monetary policy uncertainty.

9 Conclusion

9.1 Summary: what the paper concludes

This paper revisits a classic question in international macro-finance: *what is the effect of U.S. monetary policy on the nominal exchange rate?* The conventional prediction is that an unexpected U.S. easing (a negative policy-rate surprise) should weaken the dollar. The paper shows that this prediction can fail in an economically important way when monetary policy communicates information.

The main conclusions are:

- (i) **A medium-horizon sign reversal during the Great Recession.** During the Great Recession period **December 16, 2008 to September 12, 2012**, expansionary U.S. **forward-guidance (FG)** surprises caused the **dollar to appreciate** against both advanced-economy and emerging-market currencies *at business-cycle frequencies*. Importantly, the appreciation occurs **after the release of FOMC minutes** (roughly three to four weeks after the meeting), not immediately on the announcement day.
- (ii) **The result is driven by forward guidance, not QE.** The anomalous appreciation is **driven by FG shocks** and is **not driven by QE/LSAP announcements**. When the paper isolates FG shocks (or excludes QE announcement days), the sign reversal is strongest.
- (iii) **A parallel structural break for equities.** Over the same Great Recession horizon, expansionary FG surprises are followed by **declines in equity prices** (e.g., the S&P 500), whereas before and after the Great Recession, easing surprises are associated with **equity increases**. Thus, the dollar and equities move in a direction consistent with a “bad news” interpretation of easing.
- (iv) **Cross-currency heterogeneity consistent with flight to safety.** During the Great Recession, a surprise U.S. easing induces **larger dollar appreciations against worse-hedge currencies**: currencies that tend to depreciate more in global bad times. This pattern is consistent with a **flight-to-safety into USD**.
- (v) **The Great Recession was a special communication regime.** The paper emphasizes that the reversal weakens/disappears after September 2012 when the Fed switches to more **state-contingent, threshold-based** guidance that communicates the *reaction function* rather than conveying a strong signal about hidden economic conditions.

- (vi) **Implication for theory: standard models miss a key channel.** The evidence implies that models used to interpret monetary-policy effects on asset prices should incorporate the **information channel** formally; otherwise, conclusions about exchange-rate and equity responses can be systematically wrong.

9.2 Why can easing strengthen the dollar in a crisis?

The intuition is that monetary policy announcements can move exchange rates through **two conceptually distinct channels**:

1) The direct (interest-rate / expected-rate-path) channel. Holding risk premia and macro expectations fixed, lower expected U.S. short rates reduce the expected return on dollar assets. This channel pushes toward a **dollar depreciation** after an easing surprise. This is the conventional textbook mechanism.

2) The information channel (policy as a signal about fundamentals). During the Great Recession, the Fed’s calendar-based FG was interpreted as revealing **worse future U.S. macroeconomic conditions** than markets previously believed. If an easing surprise is taken as “the Fed sees something bad,” then investors revise:

- expected growth downward,
- expected inflation downward (lower expected U.S. inflation is explicitly part of the paper’s narrative),
- and effective risk aversion upward (a crisis-time risk-premium effect).

Higher risk aversion and “bad-news” revisions trigger a **flight to safety into USD assets**, which can *dominate* the direct interest-rate channel at medium horizons. This yields the observed pattern:

$$\begin{aligned}\text{easing FG news} &\Rightarrow \text{bad macro news} + \text{higher risk premia} \\ &\Rightarrow \text{USD appreciates (flight to safety) and equities fall.}\end{aligned}$$

9.3 Why the effect appears with a delay (the role of minutes)

A distinctive feature is the timing: the dollar appreciation occurs after the release of FOMC minutes. The paper’s model rationalizes this delay with an informational friction: market participants do not perfectly know the precision of the Fed’s private information at the announcement. As minutes are released, investors update about the **informativeness** of the Fed’s signal, so the information channel becomes stronger over subsequent weeks. This produces:

- a more conventional immediate response (direct channel),
- followed by a medium-horizon reversal (information + risk-premium channel).

9.4 Takeaway

During the Great Recession, U.S. forward-guidance easings were interpreted as negative information about future fundamentals, raising risk aversion and lowering expected inflation, which triggered a flight-to-safety into dollars and caused the USD to appreciate (and equities to fall) at medium horizons, especially after FOMC minutes were released and particularly against bad-hedge currencies.